



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF INFORMATION TECHNOLOGY LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015**

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INDEX

S.No	Date	Title
1.	21/01/26	Azure Devops Environment Setup.
2.	28/01/26	Azure Devops Project Setup and User Story Management.
3.	04/02/26	Setting Up Epics, Features, And User Stories for Project Planning.
4.	11/02/26	Sprint Planning.
5.	18/02/26	Poker Estimation.
6.	25/02/26	Designing Class and Sequence Diagrams for Project Architecture.
7.	11/03/26	Designing Architectural and ER Diagrams for Project Structure.
8.	18/03/26	Testing – Test Plans and Test Cases.
9.	25/03/26	Load Testing and Pipelines.
10.	08/04/26	GitHub: Project Structure & Naming Conventions.

GITHUB QR-Bloggonut



EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

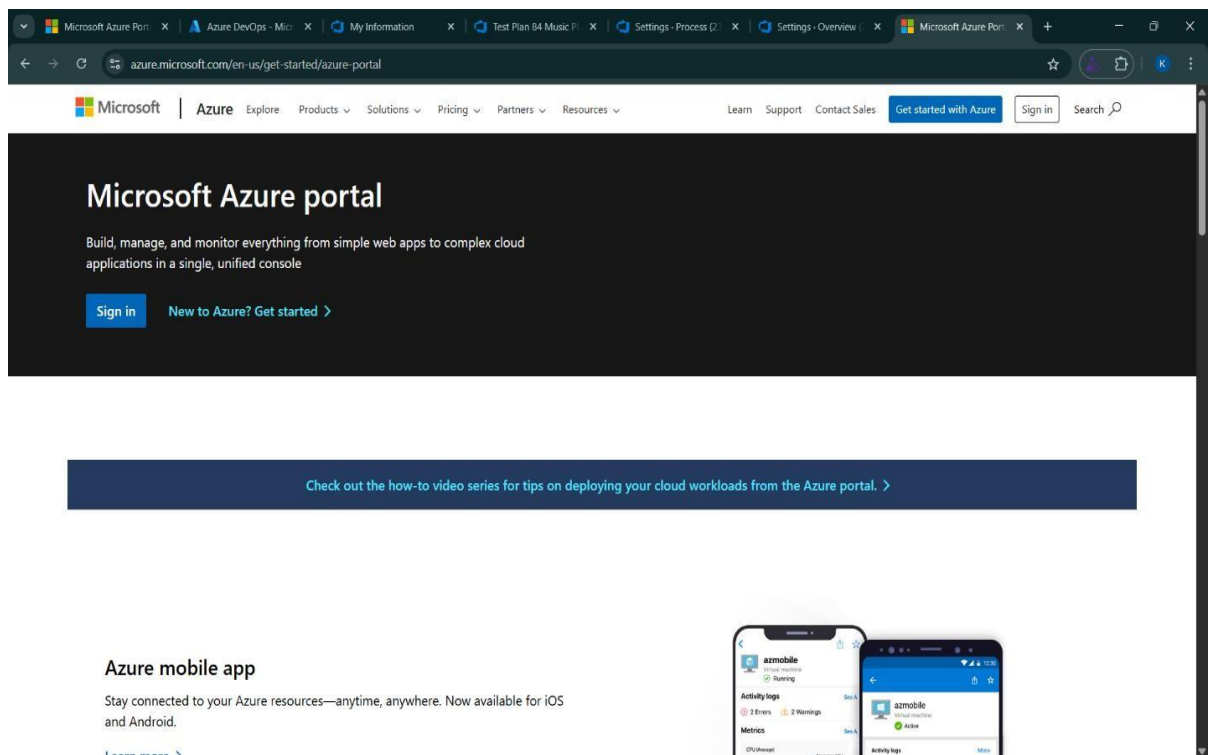
INSTALLATION:

1. Open your web browser and go to the Azure website:

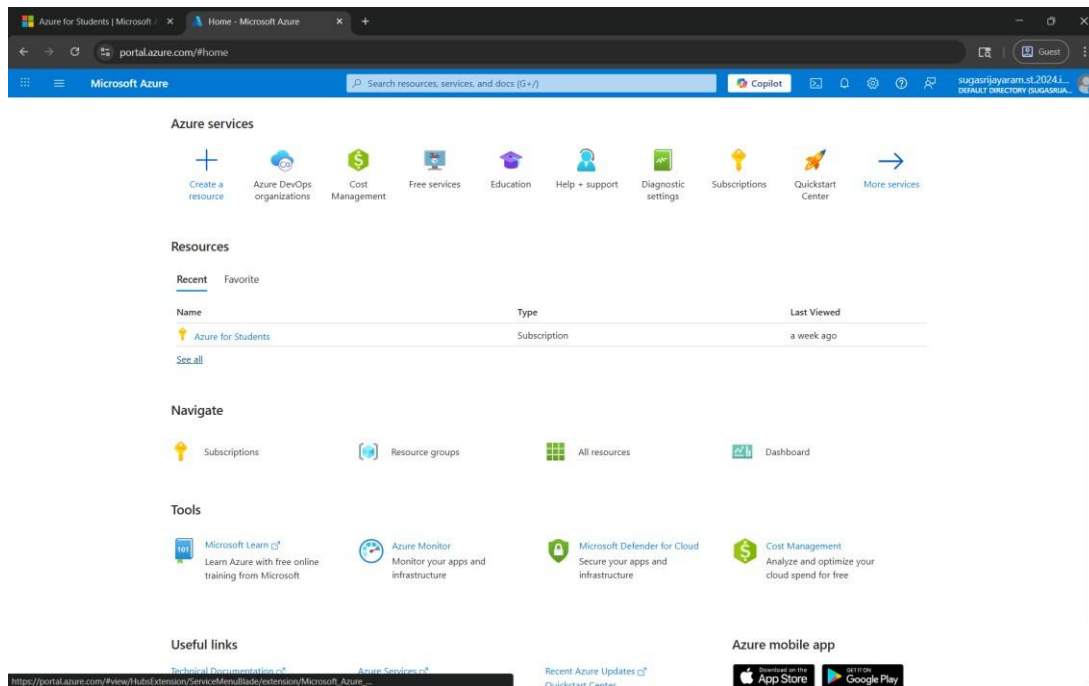
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

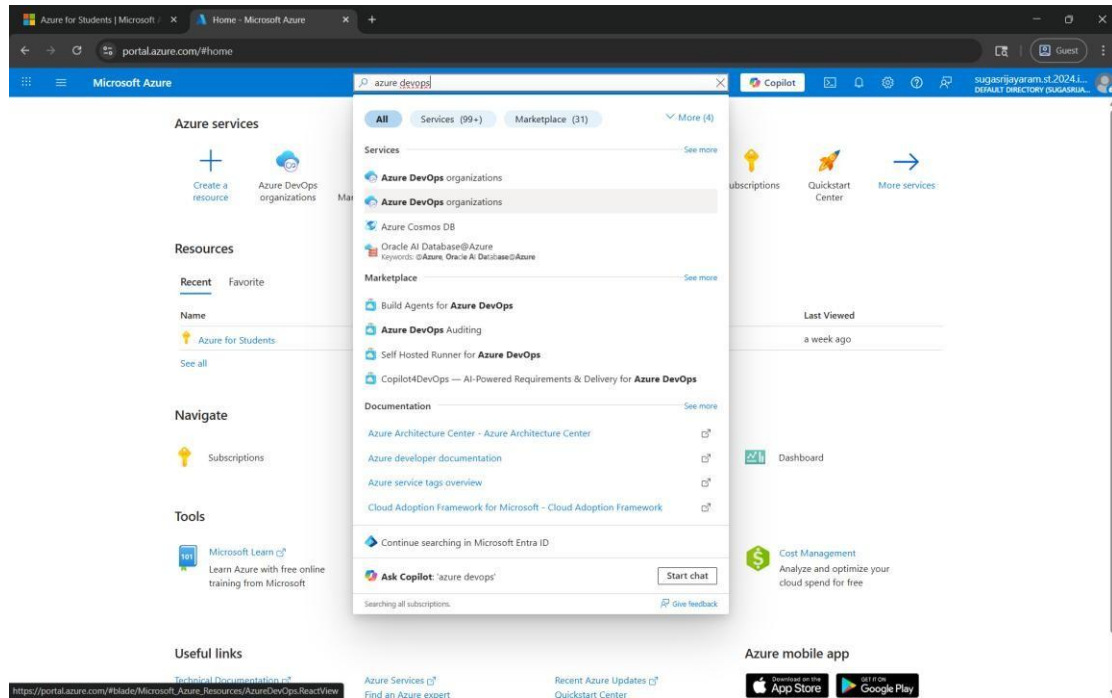
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



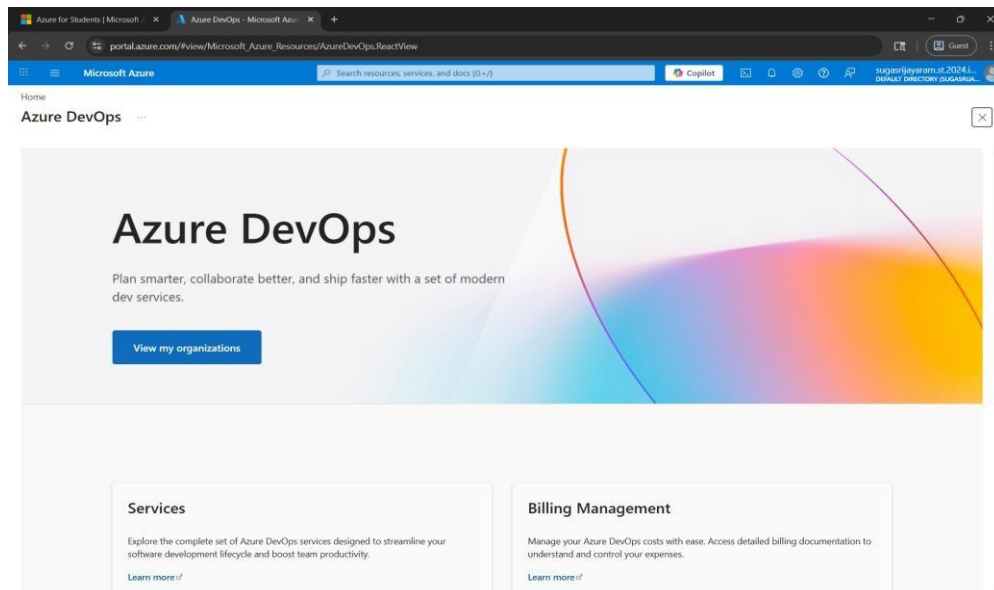
2. Azure home page

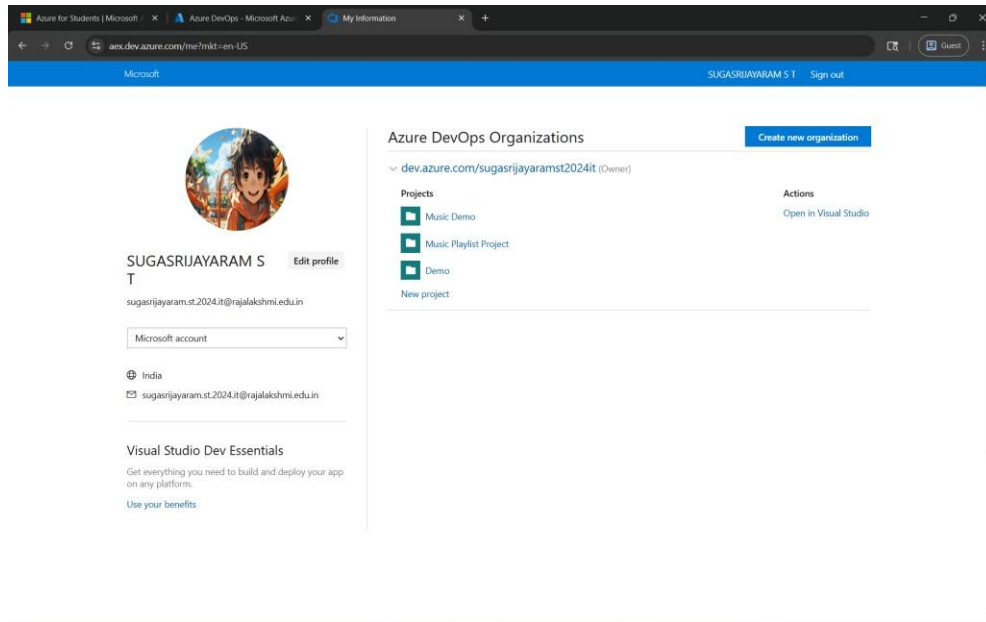


3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the **View My Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.





Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

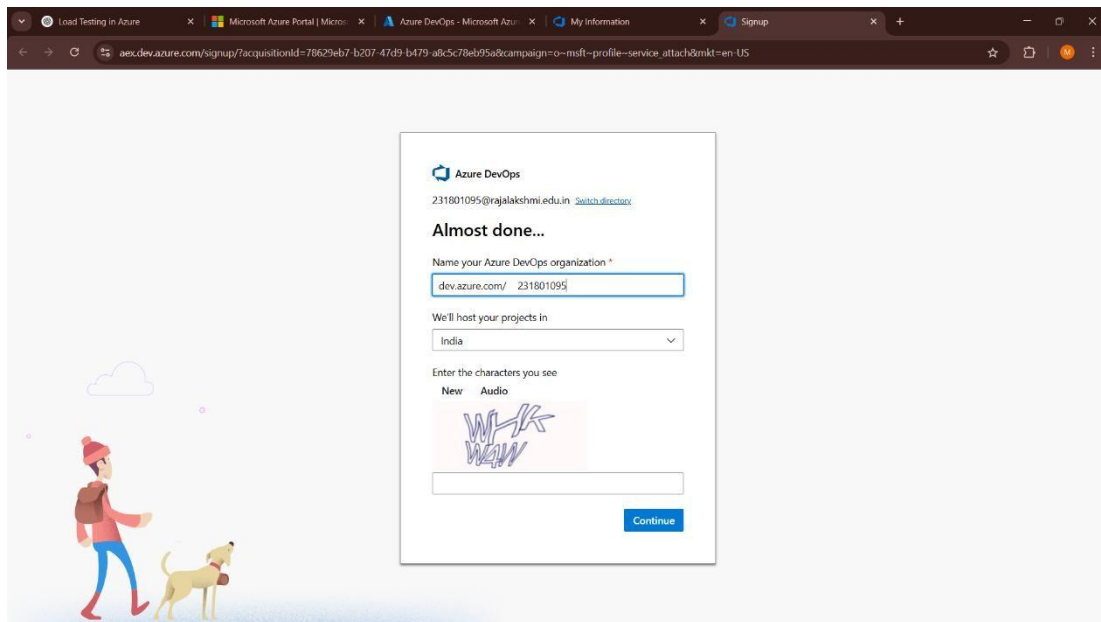
EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

Procedure:

1.Create An Azure Account



2.Create the First Project in Your Organization

- a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- b. On the organization's **Home page**, click on the **New Project** button.
- c. Enter the project name, description, and visibility options:

Name: Blog Management Project.

Description:

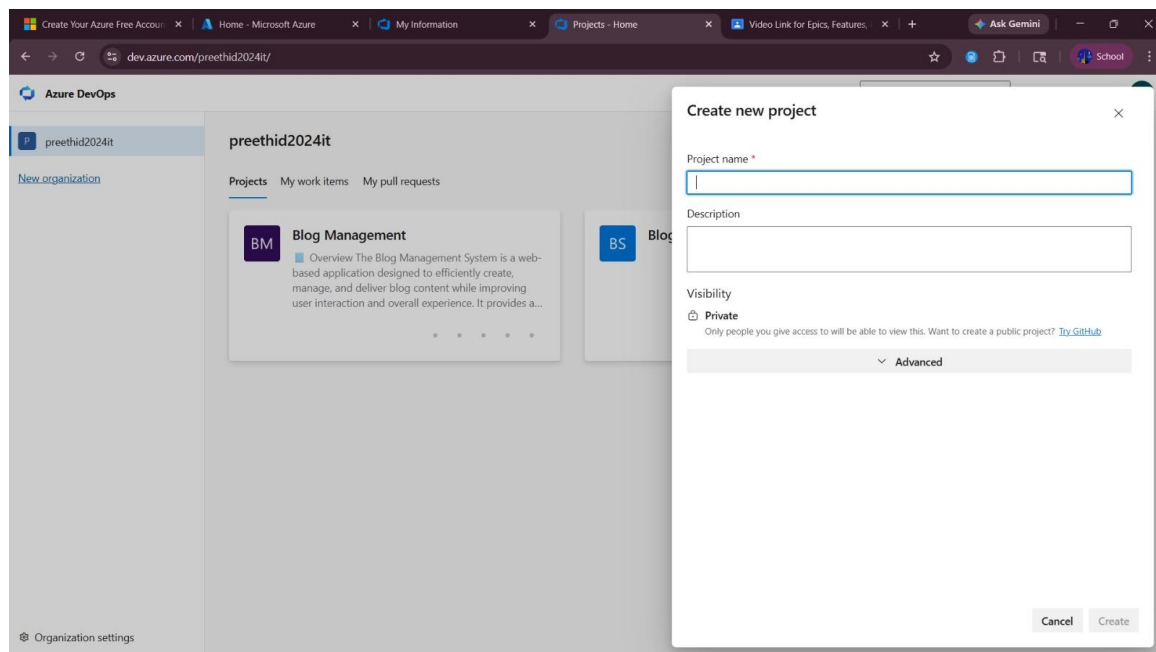
The Blog Management System is an application that allows users to create, edit, organize, and publish blog posts efficiently. Users can manage categories, upload

images, schedule posts, and interact through comments. The system also provides user authentication, content management features, and analytics to improve blog performance and reader engagement.

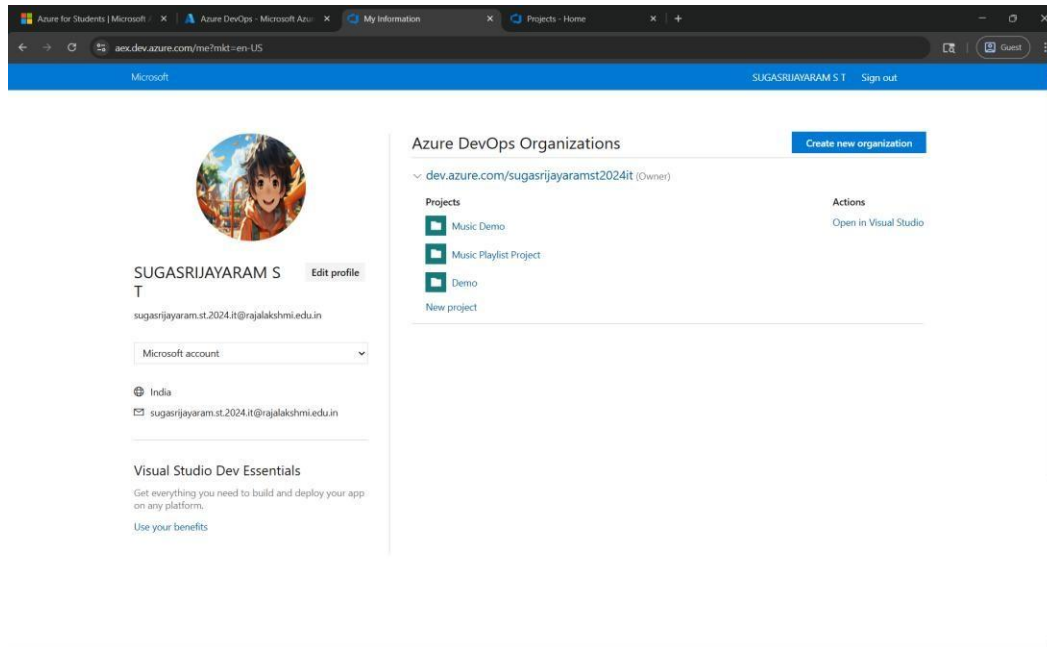
Visibility:

Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

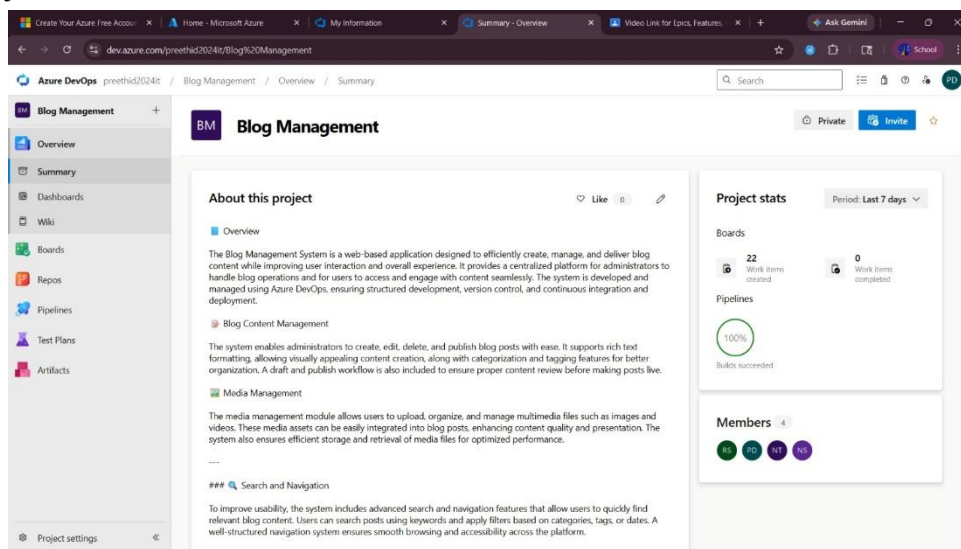
d. Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

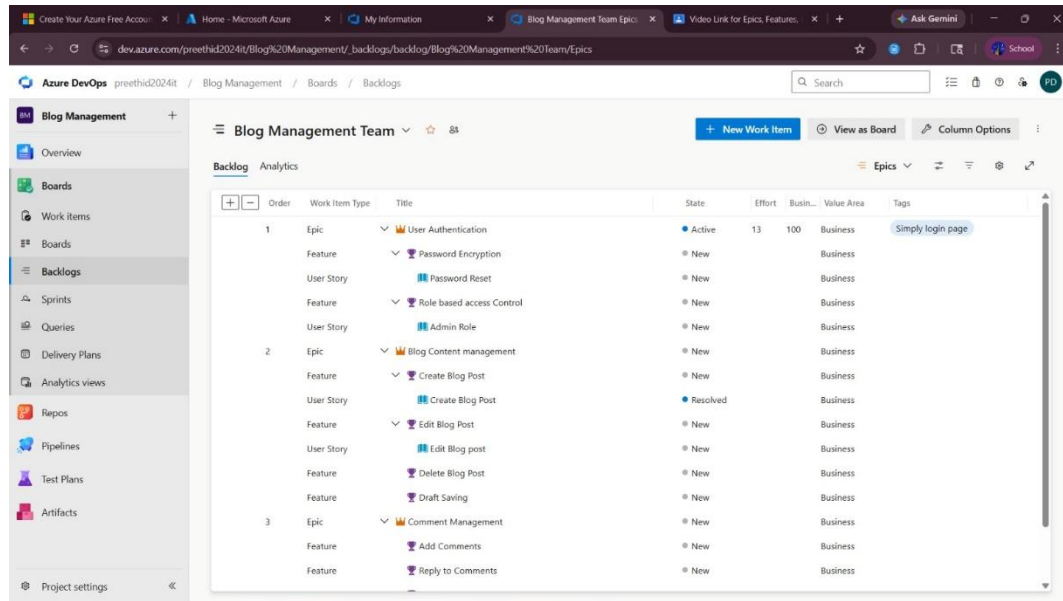


4. Project dashboard



5. To manage user stories:

- From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.
- On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for Blog Management Project.

Create Epic, Features, User Stories, Task:

The screenshot displays the Azure DevOps interface for the 'Blog Management Team' backlog. The left sidebar shows the navigation menu with 'Backlogs' selected. The main area shows a list of work items organized by Order, Work Item Type, Title, State, Effort, Business Value, and Tags.

Order	Work Item Type	Title	State	Effort	Business Value	Tags
1	Epic	User Authentication	Active	13	100	Business
	Feature	Password Encryption	New			Business
	User Story	Password Reset	New			Business
	Feature	Role based access Control	New			Business
	User Story	Admin Role	New			Business
2	Epic	Blog Content management	New			Business
	Feature	Create Blog Post	New			Business
	User Story	Create Blog Post	Resolved			Business
	Feature	Edit Blog Post	New			Business
	User Story	Edit Blog post	New			Business
	Feature	Delete Blog Post	New			Business
	Feature	Draft Saving	New			Business
3	Epic	Comment Management	New			Business
	Feature	Add Comments	New			Business
	Feature	Reply to Comments	New			Business

1.Fill in Epics

The screenshot shows the Azure DevOps interface for a work item titled "15 User registration". The work item is assigned to "Preethi D" and is in the "New" state. The description, planning, and deployment sections are visible. The description states: "Allows new users to create an account by providing necessary details such as username, email, and password. The system validates input and securely stores user credentials." The planning section shows a priority of 2 and a risk level. The deployment section provides instructions on how to track releases associated with this work item.

preethid2024it / Blog Management / Boards / Work items

Recently updated | Back to Work Items

41 of 44

FEATURE 15

15 User registration

Preethi D | 0 Comments | Add Tag

Save | Follow | Details | 1 | 0

Updated by Preethi D: Just now

State: New | Area: Blog Management | Reason: New | Iteration: Blog Management

Description

Allows new users to create an account by providing necessary details such as username, email, and password. The system validates input and securely stores user credentials.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Priority: 2

Risk:

Effort:

Business Value:

Time Criticality:

Start Date: Select a date...

Target Date: Select a date...

Classification

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Add an existing work item as a parent

2.Fill in Features

The screenshot shows the Azure DevOps interface for a work item titled "15 User registration". The work item is assigned to "Preethi D" and is in the "New" state. The description, planning, and deployment sections are visible. The description states: "Allows new users to create an account by providing necessary details such as username, email, and password. The system validates input and securely stores user credentials." The planning section shows a priority of 2 and a risk level. The deployment section provides instructions on how to track releases associated with this work item.

preethid2024it / Blog Management / Boards / Work items

Recently updated | Back to Work Items

41 of 44

FEATURE 15

15 User registration

Preethi D | 0 Comments | Add Tag

Save | Follow | Details | 1 | 0

Updated by Preethi D: Just now

State: New | Area: Blog Management | Reason: New | Iteration: Blog Management

Description

Allows new users to create an account by providing necessary details such as username, email, and password. The system validates input and securely stores user credentials.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Priority: 2

Risk:

Effort:

Business Value:

Time Criticality:

Start Date: Select a date...

Target Date: Select a date...

Classification

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

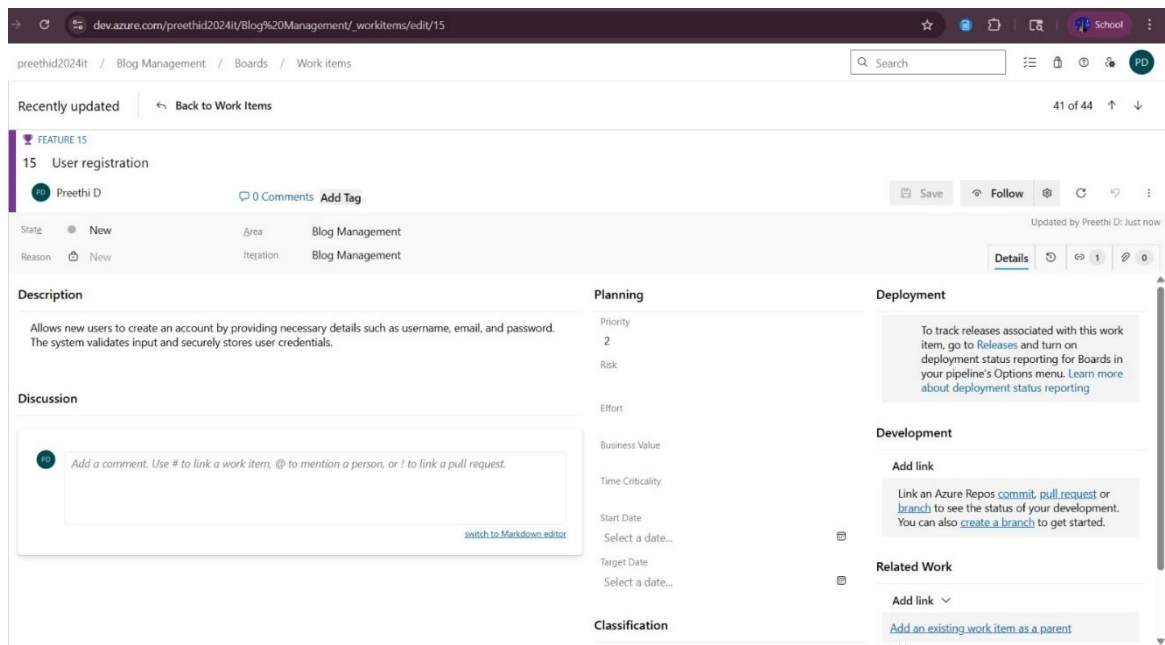
Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Add an existing work item as a parent

3.Fill in User Story Details



Result:

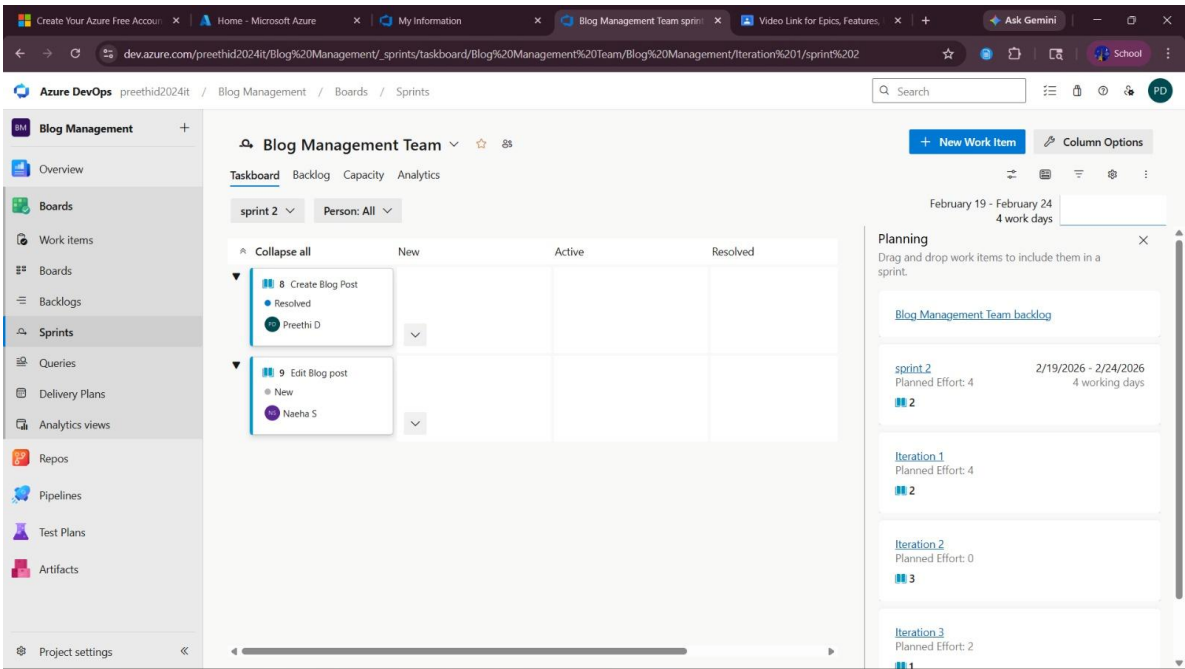
Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4	SPRINT PLANNING
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Aim:
To assign user story to specific sprint for the Blog Management Project.

Sprint Planning:

Sprint 1



Sprint 2

dev.azure.com/preethid2024it/Blog%20Management/_sprints/taskboard/Blog%20Management%20Team/Blog%20Management/Iteration%201

Azure DevOps preethid2024it / Blog Management / Boards / Sprints

Blog Management Team

Taskboard Backlog Capacity Analytics

Iteration 1 Person: All

Collapse all	New	Active	Resolved
8 Create Blog Post Resolved Preethi D			
9 Edit Blog post New Neetha S			

Planning

Drag and drop work items to include them in a sprint.

Blog Management Team backlog

sprint 2
Planned Effort: 4
2/19/2026 - 2/24/2026
4 working days
2

Iteration 1
Planned Effort: 4
2

Iteration 2
Planned Effort: 0
3

Iteration 3
Planned Effort: 2
1

Sprint 3

dev.azure.com/preethid2024it/Blog%20Management/_sprints/taskboard/Blog%20Management%20Team/Blog%20Management/Iteration%202

Azure DevOps preethid2024it / Blog Management / Boards / Sprints

Blog Management Team

Taskboard Backlog Capacity Analytics

Iteration 2 Person: All

Collapse all	New	Active	Resolved
41 Register New Unassigned			
42 Password Reset New Unassigned			
43 Admin Role New Unassigned			

Planning

Drag and drop work items to include them in a sprint.

Blog Management Team backlog

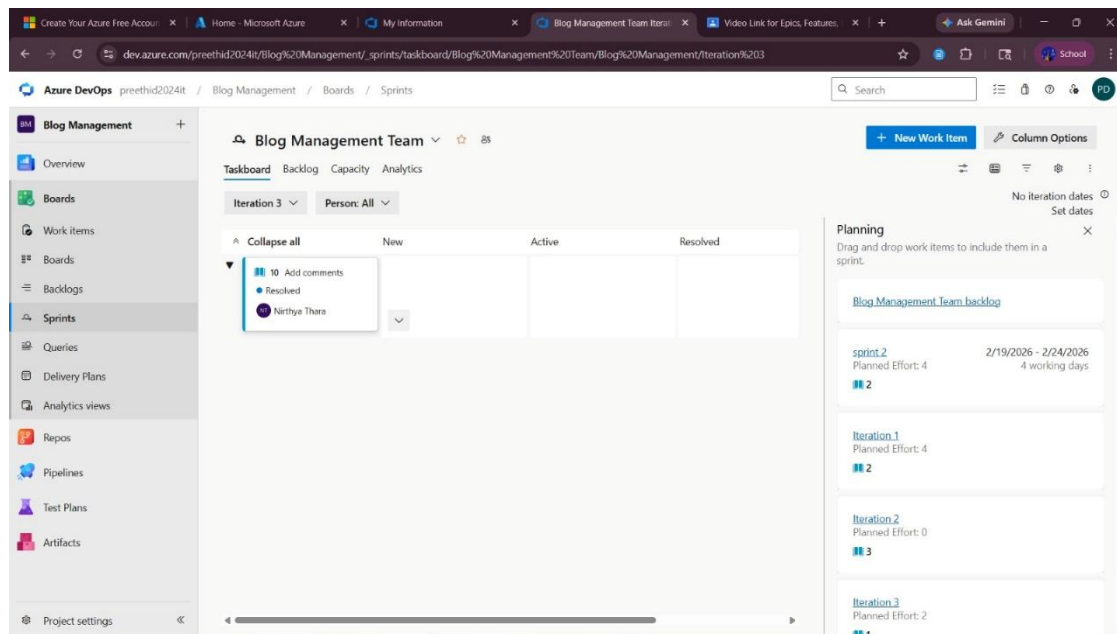
sprint 2
Planned Effort: 4
2/19/2026 - 2/24/2026
4 working days
2

Iteration 1
Planned Effort: 4
2

Iteration 2
Planned Effort: 0
3

Iteration 3
Planned Effort: 2
1

Sprint 4



Result:

The Sprints are created for the Blog Management Project.

EXP NO: 5	POKER ESTIMATION
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Aim:

Create Poker Estimation for the user stories for Blog Management Project.

Planning Poker Estimation:

User Story ID: 20

Title: User Registration and Authentication

Description:

As a user, I want to register and log in securely so that I can create, manage, and access my personal blog posts and account information.

Story Point Estimation: 8 Points

Team Poker Votes:

Team Member	Vote (Story Points)
Team Member 1	5
Team Member 2	8
Team Member 3	8
Team Member 4	8

Final Consensus: 8 Points

Reason for Estimation:

- Includes multiple features: registration, login, session, logout
- Requires frontend + backend + database integration
- Security implementation (password hashing, authentication)
- Moderate complexity with high risk

Risks:

- Security vulnerabilities
- Session management issues
- Integration challenges

Acceptance Criteria Covered:

- User registration with email & password
- Login with valid credentials
- Error message for invalid login
- Session maintained after login
- Logout functionality

Conclusion:

Final agreed estimation is 8 story points based on complexity, effort, and risk.

The screenshot shows an Azure DevOps work item interface. The work item is titled "20 User Registration and Authentication" and is in the "Resolved" state. It is assigned to "SUGASRIJAYARAM S T" and has 0 comments. The work item is part of the "Music Playlist Project" and is in the "Music Playlist Project\Iteration 1" iteration. The description is "As a user, I want to register and log in securely so that I can access my personal music playlists and downloads." The acceptance criteria are:

- User can register using email and password
- User can log in with valid credentials
- Invalid credentials show error message
- User session is maintained after login
- Logout functionality is available

 The planning section shows 8 story points, priority 1, and risk 1 - High. The classification is Business. The deployment section has a link to track releases. The development section has a link to add a link. The related work section shows a list of related work items:

- Parent: 14 User Management (Updated 9 Mar @ New)
- Child: 28 Create user database structure (Updated 11 Mar @ New)
- Child: 29 Implement login and logout logic (Updated 11 Mar @ New)
- Child: 27 Implement registration functionality (Updated 11 Mar @ New)

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

EXP NO: 6	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE
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Aim:

To Design a Class Diagram and Sequence Diagram for the Blog Management Project.

Class Diagram:

Classes:

- User
- BlogPost
- Category
- Comment
- Tag
- Payment
- Subscription
- Transaction
- Feedback
- Notification
- Advertisement

Key Attributes:

User:

name, email, role, isPremium

BlogPost:

title, content, publishDate, status

Category:

name, description

Comment:

commentText, commentDate

Tag:

tagName

Subscription:

planType, expiryDate

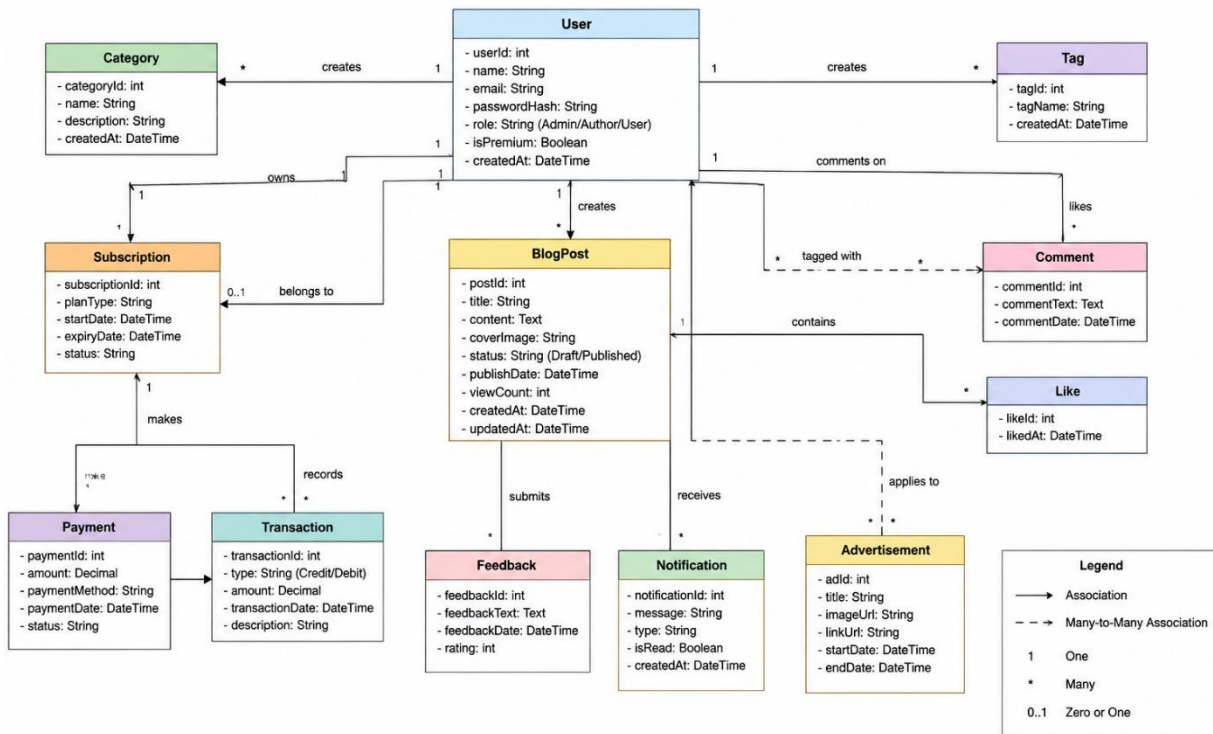
Transaction:

type, amount

Relationships:

- User → creates → BlogPost
- BlogPost → belongs to → Category
- BlogPost → contains → Comment
- BlogPost → tagged with → Tag
- User → comments on → BlogPost
- User → likes → BlogPost
- User → owns → Subscription
- Subscription → records → Transaction
- User → makes → Payment
- User → submits → Feedback
- User → receives → Notification
- Advertisement → applies to → BlogPost/Category

Blog Management System – Class Diagram



Sequence Diagram:

Actors:

User → Frontend → Backend → Database → Auth Service

Steps:

1. User enters email and password
2. Frontend validates input (email format, password)
3. Frontend sends login request to backend
4. Backend validates request data
5. Backend checks user in database

Condition: User Not Found

- Backend returns error: Invalid credentials

Condition: User Found

6. Backend verifies password
7. If password incorrect → return error
8. If password correct → proceed
9. Backend verifies user with auth service
10. Backend generates JWT token
11. Backend sets session (token expiry)

12. Backend sends success response
13. Frontend stores token
14. Frontend redirects to blog dashboard
15. User login successful



Result: The Class Diagram and Sequence Diagram is designed Successfully for the Blog Management Project.

EXP NO: 7	DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE
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Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

Architectural Diagram:

1. Client Layer

- Browser
 - React App
 - Redux / Zustand (State Management)
-

2. API Communication Layer

- Axios (HTTP Requests)
 - REST API Calls
-

3. Backend Layer

- Express Server
 - Routes
 - Controllers
 - Middleware
-

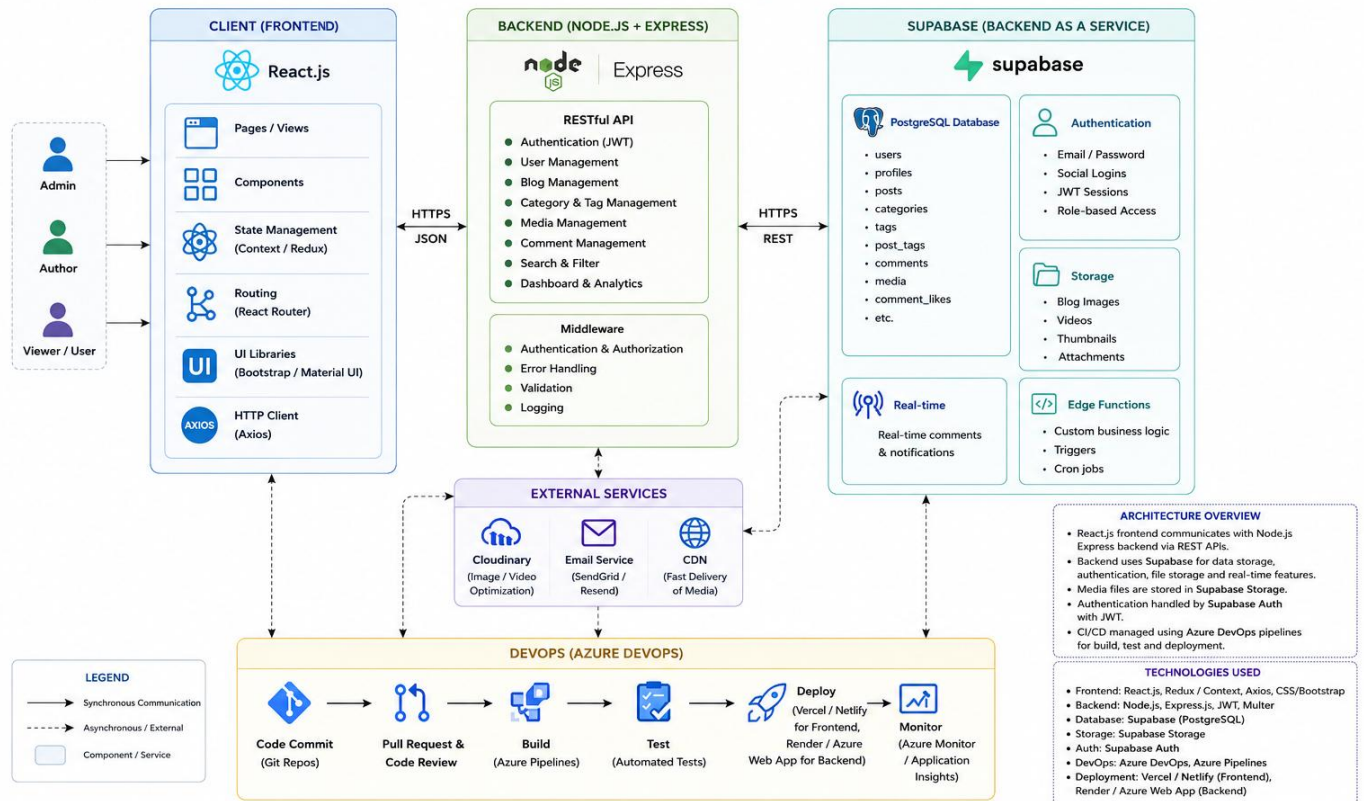
4. Services Layer

- Authentication Service (JWT / Clerk / Firebase Auth)
 - Image Upload Service (Cloudinary)
 - Notification Service
 - Email Service
-

5. Database Layer

- Mongoose Models
- MongoDB

Blog Management System Architecture MERN Stack with Supabase



ER Diagram:

Entities & Attributes:

User

- userId
- fullName
- email
- password
- role
- isPremium
- subscriptionPlan
- subscriptionExpiry

BlogPost

- postId
 - title
 - content
 - categoryId
 - authorId
 - publishDate
 - status
 - coverImage
-

Category

- categoryId
 - categoryName
 - description
-

Comment

- commentId
 - userId
 - postId
 - commentText
 - commentDate
-

Tag

- tagId
 - tagName
-

Like

- likeId
 - userId
 - postId
-

Subscription

- subscriptionId
- userId
- planType

- startDate
 - expiryDate
 - status
-

Transaction

- transactionId
 - amount
 - type
 - transactionDate
-

Payment

- paymentId
 - userId
 - amount
 - status
 - paymentMethod
-

Feedback

- feedbackId
 - userId
 - message
 - rating
-

Notification

- notificationId
 - userId
 - message
 - type
-

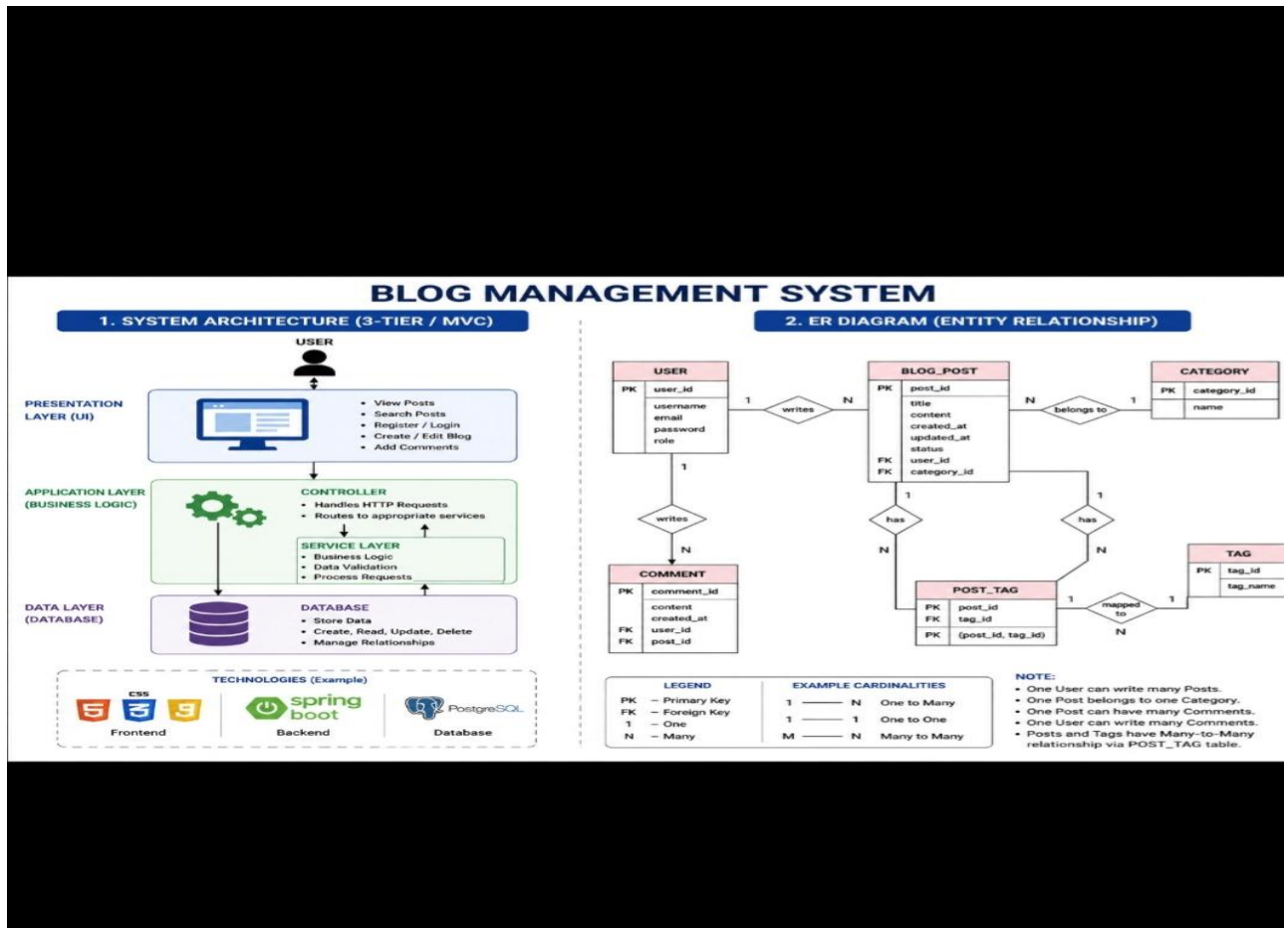
Advertisement

- advertisementId
- title
- content
- targetCategory

- startDate
 - endDate
-

Relationships

- User creates BlogPost (1 : M)
- BlogPost belongs to Category (M : 1)
- BlogPost contains Comment (1 : M)
- BlogPost tagged with Tag (M : N)
- User comments on BlogPost (1 : M)
- User likes BlogPost (M : N)
- User owns Subscription (1 : 1)
- Subscription records Transaction (1 : M)
- User makes Payment (1 : M)
- User submits Feedback (1 : M)
- User receives Notification (1 : M)
- Advertisement applies to BlogPost / Category
- Tag connected to BlogPost (M : N)



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Blog Management Project.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
------------------	--

Aim:

Test Plans and Test Case and write two test cases for at least two user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case:

Test Case Design Procedure

1. Understand Core Features of the Application

- o User Signup & Login
- o Creating and Managing Blog Posts
- o Viewing Categories and Tags
- o Editing and Publishing Blog Posts
- o Commenting and Liking Blog Posts
- o Managing User Profiles and Subscriptions

2. Define User Interactions

- o Each test case simulates a real user behaviour (e.g., logging in, creating a blog post, adding a comment).

3. Design Happy Path Test Cases

- o Focused on validating that all features function as expected under normal conditions.
- o Example: User logs in successfully, creates a blog post, publishes content, or comments on a post.

4. Design Error Path Test Cases

- o Simulate negative or unexpected scenarios to test robustness and error handling.
- o Example: Login fails with invalid credentials, blog post save fails due to network issue, empty comment submission.

5. Break Down Steps and Expected Results

- o Each test case contains step-by-step actions and a corresponding expected outcome.
- o Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- o Test cases are named clearly (e.g., TC01 – Successful Signup, TC06 – Create and Publish Blog Post).
 - o Helps in quick identification and linking to user stories or features.
-

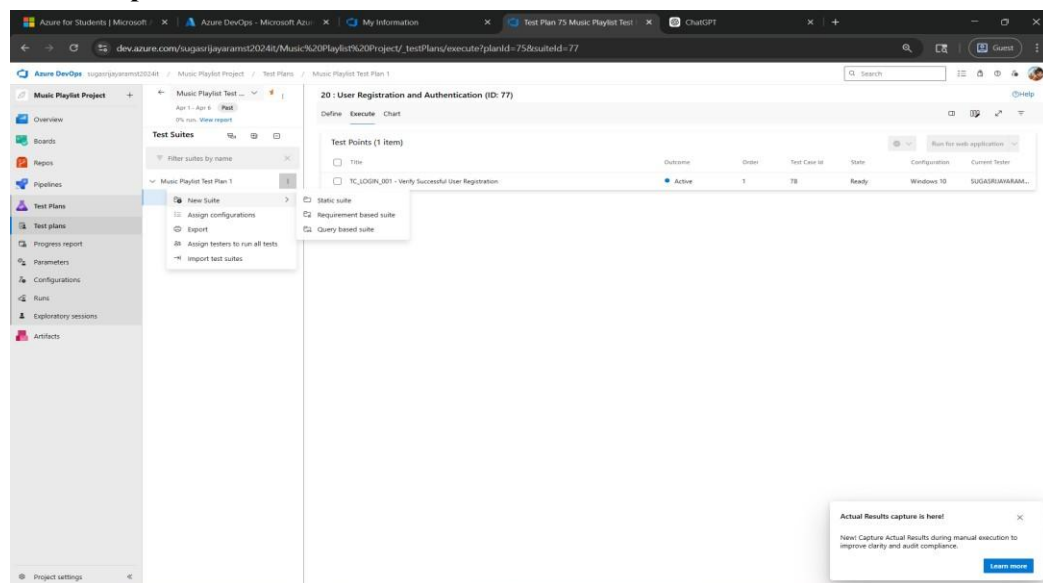
7. Separate Test Suites

- o Grouped test cases based on functionality (e.g., Login, Blog Management, Comments, Notifications).
- o Improves organization and test execution flow in Azure DevOps.

8. Prioritize and Review

- o Critical user actions are marked high-priority.
- o Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test case

USER STORIES

- As a user, I want to register and log in securely so that I can access and manage my blog account. (ID: 20)
- As a user, I want to create and manage blog posts so that I can publish and organize my content easily. (ID: 23)

Test Suites

Test Suite: TS01 - User Registration and Authentication (ID: 20)

1. TC01 – User Registration Successfully w/o Error

- Action:

- o Go to the Sign-Up page
- o Enter valid name, email, and password
- o Click "Sign Up"

- Expected Results:

- o Sign-Up form is displayed
- o Fields accept values without error
- o Account is created successfully
- o User is redirected to dashboard

- Type: Happy Path

2. TC02 – User Registration Failed with Invalid User Details

- Action:

- o Go to the Sign-Up page
- o Enter invalid details (empty fields / wrong email format / weak password)
- o Click "Sign Up"

- Expected Results:

- o Validation errors are displayed
- o Account is not created

- Type: Error Path

3. TC03 – User Login Successfully with Valid Credentials

- Action:

- o Go to Login page
- o Enter valid email and password
- o Click "Login"

- Expected Results:

- o Login form is displayed
- o Credentials are accepted
- o User is redirected to dashboard

- Type: Happy Path

4. TC04 – Login with Invalid Credentials

- Action:

- o Go to Login page
- o Enter invalid email or password
- o Click "Login"

- Expected Results:

- o Input is accepted
 - o Error message "Invalid username or password" is shown
 - Type: Error Path
-

Test Suite: TS02 - Blog Creation and Management (ID: 23)

1. TC05 – Create Blog Post Successfully

- Action:
 - o Login with valid credentials
 - o Navigate to "My Blogs"
 - o Click "Create Blog"
 - o Enter blog title and content
 - o Click "Publish/Save"
 - Expected Results:
 - o Blog post is created successfully
 - o Blog appears in the blog list
 - Type: Happy Path
-

2. TC06 – Edit Blog Post Successfully

- Action:
 - o Open an existing blog post
 - o Click "Edit"
 - o Update title or content
 - o Click "Save Changes"
 - Expected Results:
 - o Blog content is updated successfully
 - o Updated changes are visible in the blog post
 - Type: Happy Path
-

3. TC07 – Create Blog Post with Empty Fields

- Action:
 - o Navigate to "Create Blog" page
 - o Leave title/content empty
 - o Click "Publish/Save"
 - Expected Results:
 - o Validation errors are displayed
 - o Blog post is not created
 - Type: Error Path
-

4. TC08 – Delete Blog Post Successfully

- Action:

- o Navigate to "My Blogs"
- o Click "Delete" on a blog post
- o Confirm deletion

- Expected Results:

- o Blog post is deleted successfully
- o Blog is removed from the list
- o Changes are saved

- Type: Happy Path

The screenshot shows the Azure DevOps web interface. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main content area is titled '20 : User Registration and Authentication (ID: 77)'. Below the title, there are tabs for 'Define', 'Execute', and 'Chart'. A table titled 'Test Cases (4 items)' is displayed, showing the following data:

Title	Order	Test Case Id	Assigned To	State
TC01 – User Registration Successfully w/o Error	1	78	SUGASRIJAVAR...	Close
TC02 – User Registration Failed with Invalid User Details	2	79	SUGASRIJAVAR...	Close
TC03 – User Login Successfully with Valid Credentials	3	80	Sridhar C.	Close
TC04-Login with invalid credentials	4	81	Sridhar C.	Close

TEST CASE 78

78 TC01 – User Registration Successfully w/o Error

SUGASRIJAYARAM S T

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1

Updated by SUGASRIJAYARAM S T: 1h ago

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	Navigate to web page	Login page loads	
2.	Click "Get Started"	Registration form appears	
3.	Enter First name and last name	Name fields accept input	
4.	Enter Username	Username field accepts input	
5.	Enter valid email address	Email field accepts input	
6.	Enter password	Password field accepts input	
7.	Click "Continue" button	Account is created	
8.	Check email and enter verification code	Verification is successful	
9.	Click "Continue"	User is redirected to Home page	

Click or type here to add a step

Recent test results

- Passed by SUGASRIJAYARAM S T with Windows 10 Completed 12:07 PM, 4/28/2026
- Failed by SUGASRIJAYARAM S T with Windows 10 Completed 12:06 PM, 4/28/2026
- Passed by SUGASRIJAYARAM S T with Windows 10 Completed 3:20 AM, 4/28/2026

View all test result history

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

20 : User Registration and Authentication

Define Execute Chart

Test Points (4 items)

Title
☒ TC01 – User Registration Successfully w/o Error
☐ TC02 – User Registration Failed with Invalid Email
☐ TC03 – User Login Successfully with Valid Credentials
☐ TC04-Login with invalid credentials

Test point ID 2 execution history

Test Case: 78 TC01 – User Registration Successfully w/o Error

Test Suite: 77 20 : User Registration and Authentication

Current Tester: SUGASRIJAYARAM S T

Configuration: Windows 10

Outcome	Test run	Run by	Time completed
Passed	9	SUGASRIJAYARAM S...	Yesterday
Failed	8	SUGASRIJAYARAM S...	Yesterday
Passed	5	sriram S	Yesterday

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

TEST CASE 81

81 TC04-Login with invalid credentials

Sridhar C 0 Comments Add Tag

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1 Updated by SUGASRIJAYARAM S T: 42m ago

Steps Summary Associated Automation 2 0

Steps

Steps	Action	Expected result	Attachments
1.	Open login page	Login page is displayed	
2.	Enter invalid username/email	Input is accepted	
3.	Enter invalid password	Input is accepted	
4.	Click "Login"	System validates credentials	
5.	Login attempt fails	Error message displayed: "Invalid credentials"	

Click or type here to add a step

Parameter values

Recent test results

Passed by with Windows 10 Completed 12:28 PM, 4/28/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

20 User Registration and Authentication Updated Yesterday Resolved

Tests

20 User Registration and Authentication

TEST CASE 85

85 TC06 – Add and Remove Songs from Playlist

SUGASRIJAYARAM S T 0 Comments Add Tag

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1 Updated by SUGASRIJAYARAM S T: 43m ago

Steps Summary Associated Automation 2 0

Steps

Steps	Action	Expected result	Attachments
1.	Open an existing playlist	Playlist details are displayed	
2.	Click "Add Song"	Song selection options appear	
3.	Select a song	Song is added to the playlist	
4.	Verify playlist	Added song is visible in the list	
5.	Click "Remove/Delete" on a song	Song is removed	
6.	Verify playlist	Removed song no longer appears	

Click or type here to add a step

Parameter values

Recent test results

Passed by with Windows 10 Completed 4:36 PM, 4/29/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

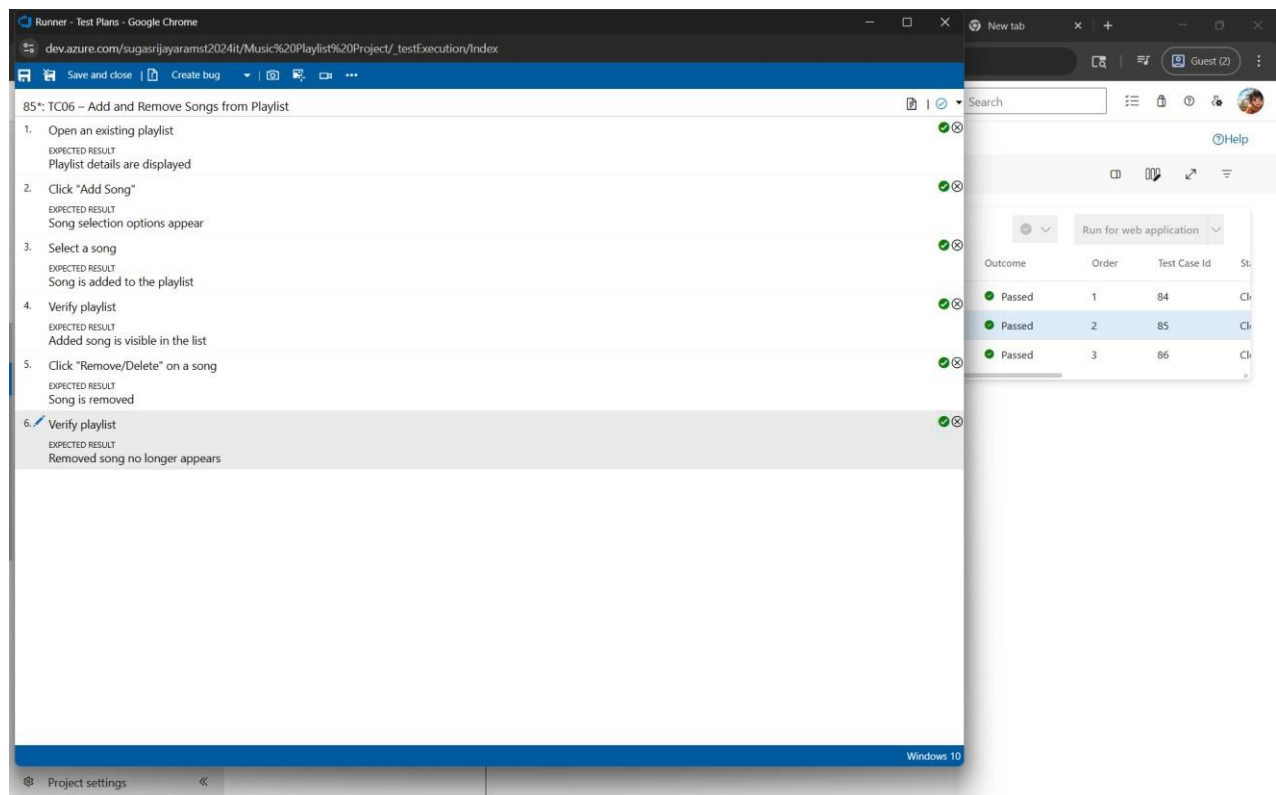
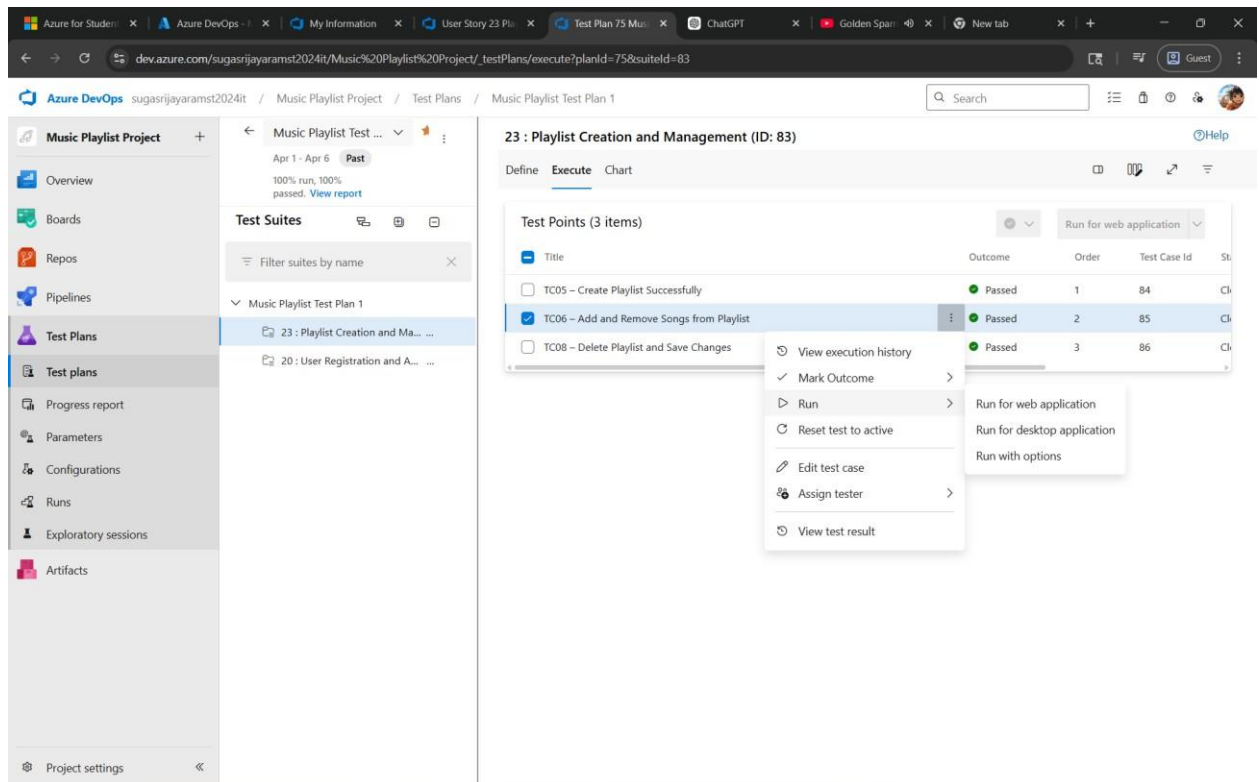
Parent

23 Playlist Creation and Management Updated 54m ago Resolved

Tests

23 Playlist Creation and Management

4. Running the test cases



5.Creating the bug

Runner - Test Plans - Google Chrome

dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testExecution/Index

Save and close | Create bug | Search

85*: TC06 - Add and Remove Songs from Playlist

- Open an existing playlist
EXPECTED RESULT
Playlist details are displayed
- Click "Add Song"
EXPECTED RESULT
Song selection options appear
COMMENT
Add notes or context
- Select a song
EXPECTED RESULT
Song is added to the playlist
COMMENT
Add notes or context
- Verify playlist
EXPECTED RESULT
Added song is visible in the list
COMMENT
Add notes or context
- Click "Remove/Delete" on a song
EXPECTED RESULT
Song is removed
COMMENT
Add notes or context
- Verify playlist
EXPECTED RESULT
Removed song no longer appears
COMMENT
Add notes or context

Run for web application

Outcome	Order	Test Case Id	St
Passed	1	84	Ch
Passed	2	85	Ch
Passed	3	86	Ch

Project settings

Runner - Test Plans - Google Chrome

dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testExecution/Index

NEW BUG *

Bug01- TC06 Add and Remove Songs from playlist

No one selected | 0 Comments | Add Tag | Save and Close

State: New | Area: Music Playlist Project | Reason: New | Iteration: Music Playlist Project/Iteration 1 | Details

Repro Steps

4/29/2026 5:27 PM Bug filed on "TC06 - Add and Remove Songs from Playlist"

Step no.	Result	Title
1.	Passed	Open an existing playlist Expected Result Playlist details are displayed
2.	Failed	Click "Add Song" Expected Result Song selection options appear
3.	Failed	Select a song Expected Result Song is added to the playlist

System Info

Click to add System Info.

Planning

Resolved Reason

Story Points

Priority
2

Severity
3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Planning Poker

Save and re-open the new work item to start estimation.

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Parent

23 Playlist Creation and Management
Updated 39m ago | Resolved

Tested By

85 TC06 - Add and Remove Songs from Playlist
Updated 49m ago | Closed

System Info

Found in Build

Integrated in Build

Discussion

6. Test case results

The screenshot shows the Azure DevOps Test Plans interface. On the left, the 'Test Suites' section shows 'Music Playlist Test Plan 1' with a sub-suite '20: User Registration and Authentication'. The main panel displays the 'Test Points (4 items)' for this suite. A modal window titled 'Test point ID 2 execution history' is open, showing a table of execution results for test point '78 TC01 - User Registration Successfully w/o Error'.

Outcome	Test run	Run by	Time completed
Passed	21	SUGASRIJAYARAM S...	19m ago
Passed	9	SUGASRIJAYARAM S...	Yesterday
Failed	8	SUGASRIJAYARAM S...	Yesterday
Passed	5	sriram S	Yesterday

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

7. Bug report

- Assigning bug to the developer and changing state to Active

The screenshot shows the Azure DevOps Bug report interface for 'Bug 87'. The bug title is 'Bug01 - TC06 Add and Remove Songs from Playlist'. The 'State' dropdown menu is open, showing options: 'Active' (selected), 'New', 'Resolved', and 'Closed'. The 'Reason' dropdown is also open, showing 'Active' (selected), 'Resolved', and 'Closed'. The 'Steps' section lists three steps, with the first two marked as 'Failed'.

Step no.	Expected Result	Actual Result
1.	Playlist details are displayed	Failed
2.	Click "Add Song"	Failed
3.	Song selection options appear	Failed

The 'System Info' section shows the bug was found in build 'Found in Build' and integrated in build 'Integrated in Build'. The 'Discussion' section is empty.

BUG 87

87 Bug01 - TC06 Add and Remove Songs from Playlist

SUGASRIJAYARAM S T 0 Comments Add Tag

State: Active Area: Music Playlist Project Reason: Regression Iteration: Music Playlist Project\Iteration 1 Updated by SUGASRIJAYARAM S T: Just now

4/29/2026 5:30 PM Bug filed on "TC06 - Add and Remove Songs from Playlist"

Step no.	Result	Title
1.	Passed	Open an existing playlist <u>Expected Result</u> Playlist details are displayed
2.	Failed	Click "Add Song" <u>Expected Result</u> Song selection options appear
3.	Failed	Select a song <u>Expected Result</u> Song is added to the playlist

System Info

Discussion

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Planning Poker

Voted: 0

0 1 2 3 5 8 13

Write your comment here

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

23 Playlist Creation and Management
Updated 58m ago Resolved

Tested By

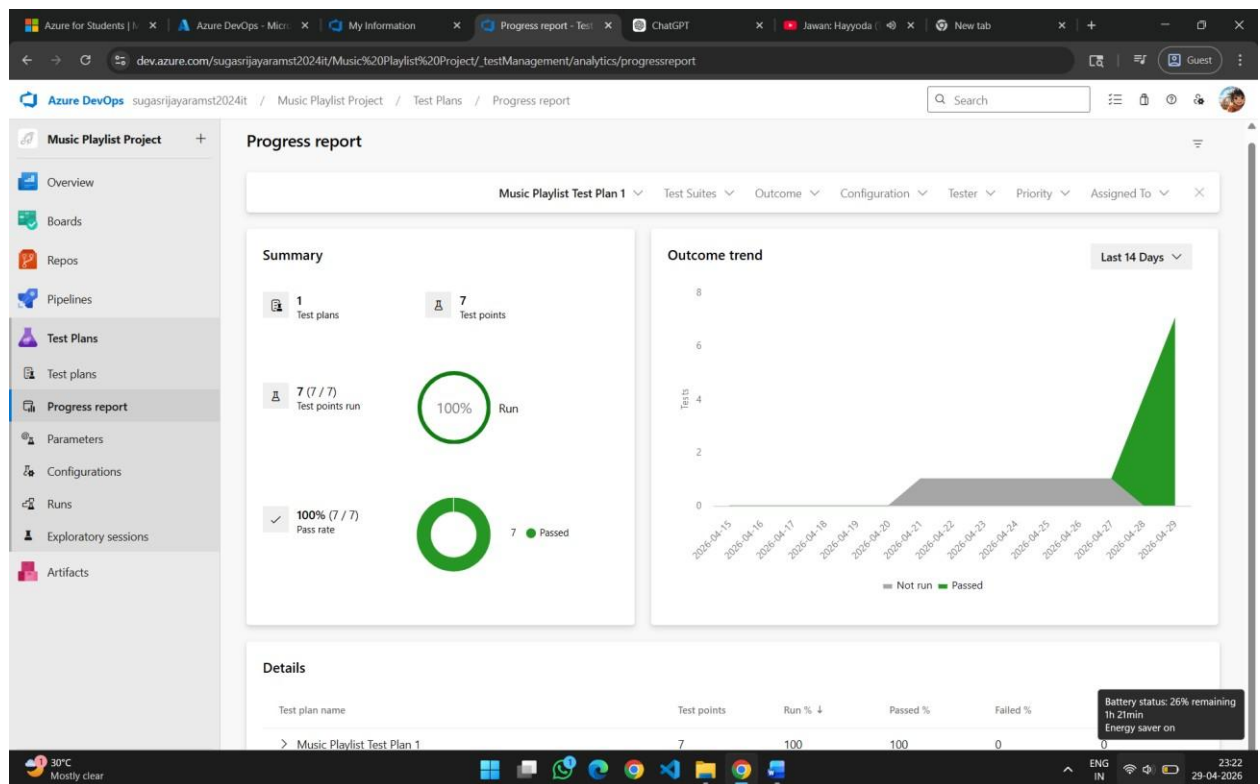
85 TC06 - Add and Remove Songs from Playlist
Updated 1h ago Closed

System Info

Found in Build

Integrated in Build

8. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.

EXP NO: 9	LOAD TESTING AND PIPELINES
-----------	-----------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

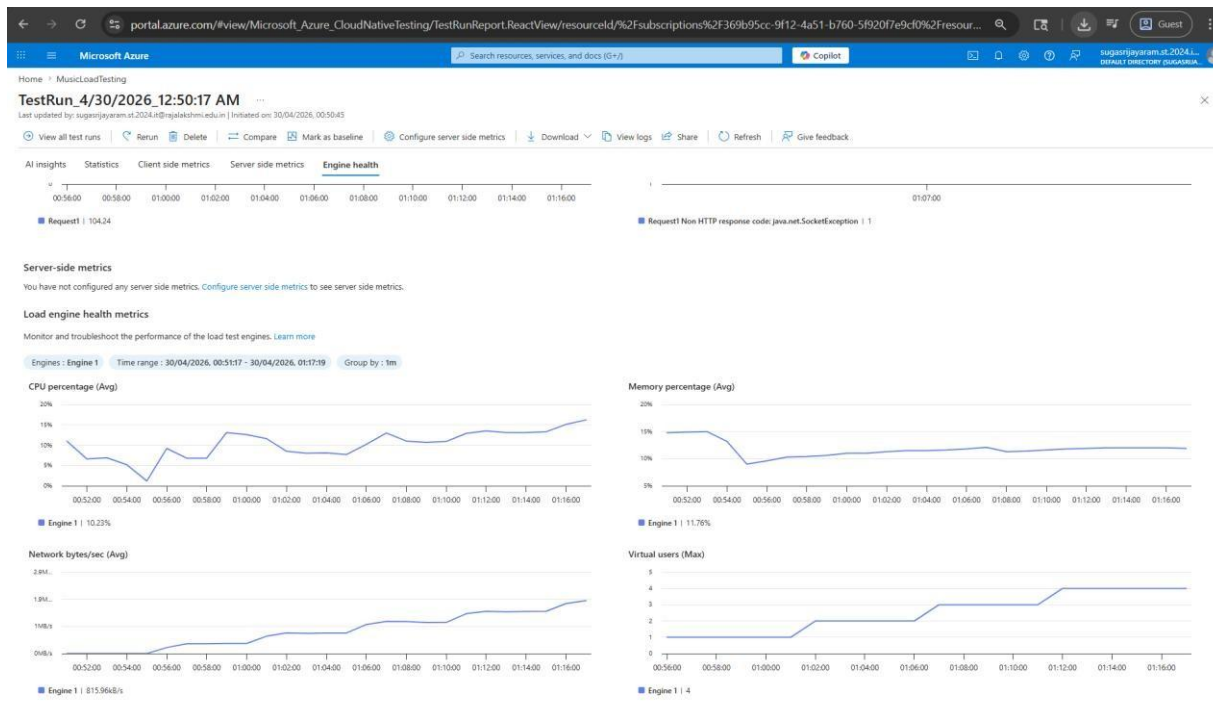
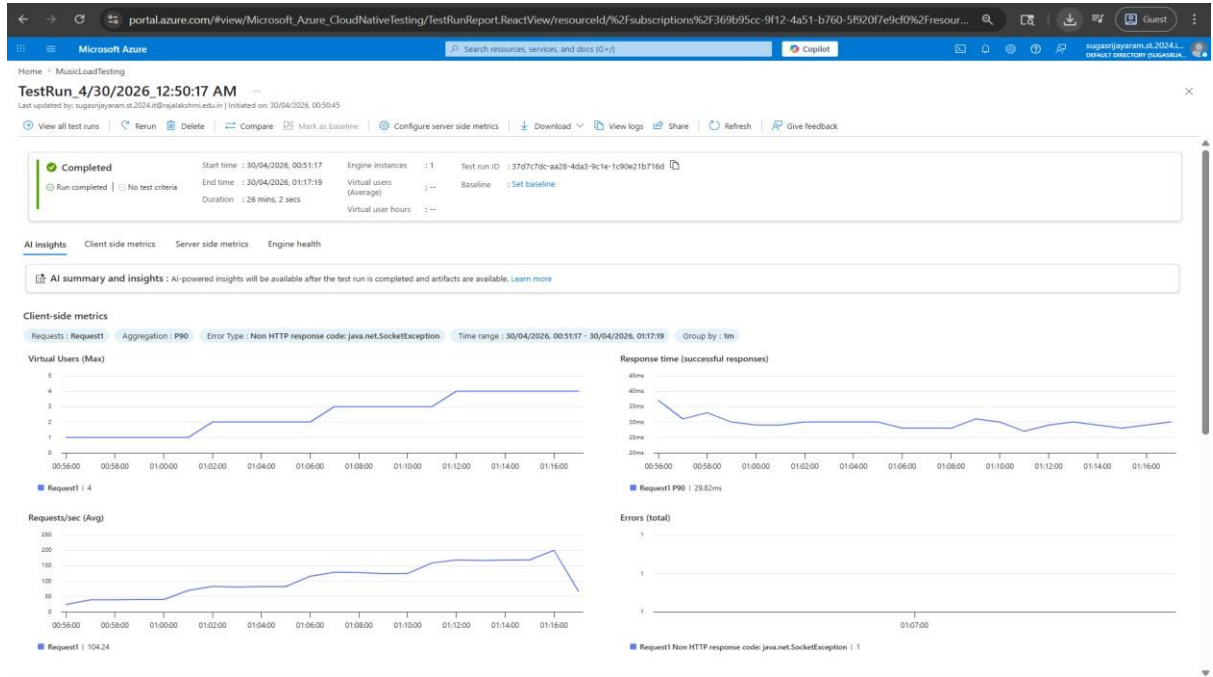
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource ○ Go to *Create a resource* → Search for “Azure Load Testing”.
○ Select Azure Load Testing and click Create.
3. Fill in the Configuration Details ○ *Subscription*: Choose your Azure subscription. ○ *Resource Group*: Create new or select an existing one. ○ *Name*: Provide a unique name (no special characters).
○ *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

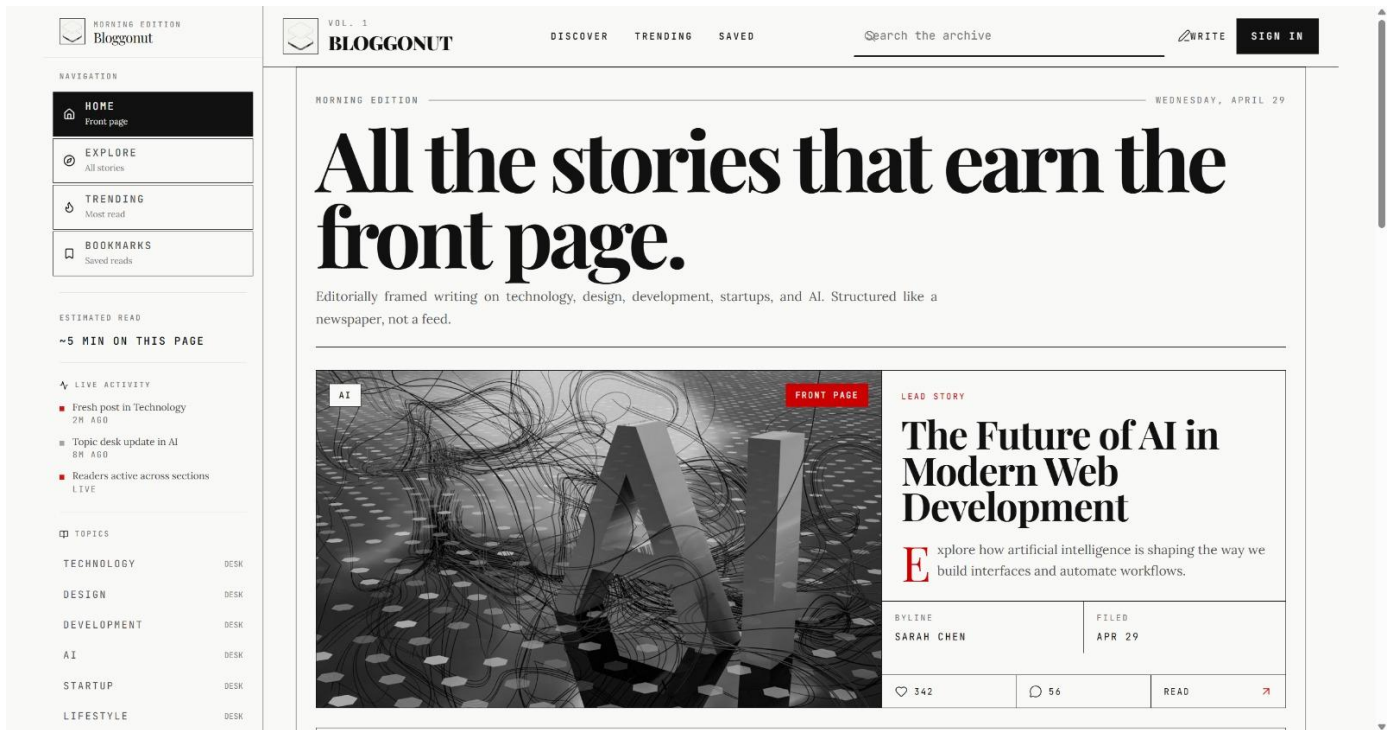
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab ○ *Test Name*: Provide a unique name.
○ *Description*: (Optional) Add test purpose.
○ *Run After Creation*: Keep checked.
3. Load Settings ○ *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yaml Code

```
trigger:  -
main

pool: vmImage: 'ubuntu-
latest'

variables:
  NODE_VERSION: '20.19.x'
  NPM_CACHE: '$(Pipeline.Workspace)/.npm'

steps:

# Install Node.js (UPDATED TASK)
- task: UseNode@1  inputs:
  version: $(NODE_VERSION)
  displayName: 'Install Node.js'

# Cache npm (FIXED PATH) -
task: Cache@2  inputs:
  key: 'npm | "$(Agent.OS)" | **/package-lock.json'
  path: $(NPM_CACHE)  displayName: 'Cache npm'

# Install Frontend Dependencies
- script: |
  npm config set cache $(NPM_CACHE) --global
  npm ci  workingDirectory: frontend
  displayName: 'Install Frontend Dependencies'

# Build Frontend (Vite) -
script: npm run build
workingDirectory: frontend
displayName: 'Build Frontend'

# Frontend Tests (optional)
- script: npm test --if-present
  workingDirectory: frontend
  displayName: 'Run Frontend Tests'

# Install Backend Dependencies
- script: |
  npm config set cache $(NPM_CACHE) --global
  npm ci  workingDirectory: backend
  displayName: 'Install Backend Dependencies'
```

```

# Backend Tests
- script: npm test --if-present
  workingDirectory: backend
  displayName: 'Run Backend Tests'

# Verify Frontend Build (fail early)
- script: | if [ ! -d "frontend/dist" ]; then
  echo "Build failed: dist folder not found"
  exit 1    fi
  displayName: 'Verify Frontend Build'

# Archive Frontend -
task: ArchiveFiles@2
inputs:
  rootFolderOrFile: 'frontend/dist'
  includeRootFolder: false  archiveType:
'zip'
  archiveFile: '$(Build.ArtifactStagingDirectory)/frontend.zip'
  displayName: 'Archive Frontend'

# Archive Backend -
task: ArchiveFiles@2
inputs:
  rootFolderOrFile: 'backend'
  includeRootFolder: false  archiveType:
'zip'
  archiveFile: '$(Build.ArtifactStagingDirectory)/backend.zip'
  displayName: 'Archive Backend'

# Publish Artifacts
- task: PublishBuildArtifacts@1
inputs:
  PathToPublish: '$(Build.ArtifactStagingDirectory)'
  ArtifactName: 'drop'  displayName: 'Publish
Artifacts'

```

3. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

Azure DevOps

preethid2024it

Blog Management

Pipelines

Naeha-S.Bloggonut

20260510.3

Q Search

NT

Blog Management

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings

Jobs in run #20260510.3

Naeha-S.Bloggonut

Jobs

Job

3s

Initialize job

<1s

Checkout Naeha-S/Bloggonut@m...

2s

Run a one-line script

<1s

Run a multi-line script

<1s

Post-job: Checkout Naeha-S/Blo...

<1s

Finalize Job

<1s

Initialize job

Q Search

View raw log

```
1 Starting: Initialize job
2 Agent name: 'Azure Pipelines 1'
3 Agent machine name: 'runnermddnds'
4 Current agent version: '4.273.0'
5 Runner Image Provisioner
12 Internal runner telemetry
15 Operating System
19 Runner Image
24 Current image version: '20260413.86.1'
25 Agent running as: 'vsts'
26 Prepare build directory.
27 Set build variables.
28 Download all required tasks.
29 Downloading task: CmdLine (2.268.0)
30 Checking job knob settings.
31 Knob: DockerActionRetries = true Source: $(VSTSAGENT_DOCKER_ACTION_RETRIES)
32 Knob: AgentToolsDirectory = /opt/hostedtoolcache Source: $(AGENT_TOOLS_DIRECTORY)
33 Knob: UseGitLongPaths = true Source: $(USE_GIT_LONG_PATHS)
34 Knob: UseNode2WithHandlerData = True Source: $(DistributedTask.Agent.UseNode2WithHandlerData)
35 Knob: EnableIssueSourceValidation = true Source: $(ENABLE_ISSUE_SOURCE_VALIDATION)
36 Knob: AgentEnablePipelineArtifactLargeChunkSize = true Source: $(AGENT_ENABLE_PIPELINEARTIFACT_LARGE_CHUNK_SIZE)
37 Knob: ContinueAfterCancelProcessTreeKillAttempt = true Source: $(VSTSAGENT_CONTINUE_AFTER_CANCEL_PROCESS_TREE_KILL_ATTEMPT)
38 Knob: ProcessHandlerSecureArguments = false Source: $(AZP_75787_ENABLE_NEW_LOGIC)
39 Knob: ProcessHandlerSecureArguments = false Source: $(AZP_75787_ENABLE_NEW_LOGIC_LOG)
40 Knob: ProcessHandlerTelemetry = true Source: $(AZP_75787_ENABLE_COLLECT)
41 Knob: UseNewModelHandlerTelemetry = True Source: $(DistributedTask.Agent.USE_NEW_MODEL_HANDLER_TELEMETRY)
42 Knob: ProcessHandlerEnableNewLogic = true Source: $(AZP_75787_ENABLE_NEW_PH_LOGIC)
43 Knob: EnableResourceMonitorDebugOutput = true Source: $(AZP_ENABLE_RESOURCE_MONITOR_DEBUG_OUTPUT)
44 Knob: EnableResourceUtilizationWarnings = true Source: $(AZP_ENABLE_RESOURCE_UTILIZATION_WARNINGS)
45 Knob: IgnoreVSTaskLib = true Source: $(AZP_AGENT_IGNORE_VSTASKLIB)
46 Knob: FailJobWhenAgentDies = true Source: $(FAIL_JOB_WHEN_AGENT_DIES)
47 Knob: EnhancedWorkerCrashHandling = true Source: $(AZP_ENHANCED_WORKER_CRASH_HANDLING)
48 Knob: CheckForTaskDeprecation = true Source: $(AZP_AGENT_CHECK_FOR_TASK_DEPRECATION)
49 Knob: CheckIfTaskNodeRunnerIsDeprecated246 = true Source: $(DistributedTask.Agent.CheckIfTaskNodeRunnerIsDeprecated246)
50 Knob: EnableTimeoutLogFlushing = True Source: $(DistributedTask.Agent.EnableTimeoutLogFlushing)
51 Knob: LogTaskNameInUserAgent = true Source: $(AZP_AGENT_LOG_TASKNAME_IN_USERAGENT)
52 Knob: UseFetchFilterInCheckoutTask = true Source: $(AGENT_USE_FETCH_FILTER_IN_CHECKOUT_TASK)
53 Knob: Rosetta2Warning = true Source: $(ROSETTA2_WARNING)
54 Knob: CheckPsModulesLocations = False Source: $(DistributedTask.Agent.CheckPsModulesLocations)
55 Knob: AddSourceCredentialInTestGitCheckoutTask = True Source: $(DistributedTask.Agent.AddSourceCredentialInTestGitCheckoutTask)
```

Azure DevOps

preethid2024it

Blog Management

Pipelines

Naeha-S.Bloggonut

20260510.3

Q Search

NT

Blog Management

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings

Pipelines

Recent

All

Runs

New pipeline

Filter pipelines

Recently run pipelines

Pipeline	Last run
Blog Management	#20260511.1 • test: add comprehensive test suite for backend controllers, middleware, and frontend utils Manually run by copilot/analyze-test-coverage 27m ago 6s
Naeha-S.Bloggonut	#20260510.3 • Merge pull request #2 from Naeha-S/copilot/analyze-test-coverage Individual CI for main yesterday 10s

Project settings

https://dev.azure.com/preethid2024it/Blog%20Management/_build?view=pipelines

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS
------------	---

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Blog Management project.

GitHub Project Structure

Bloggonut Public

Watch 0 Fork 0 Star 0

main 3 Branches 0 Tags

Go to file Add file Code

NirthyThara Add project details to README.md b780b8 · 1 minute ago 14 Commits

File/Folder	Description	Last Commit
backend	test: add comprehensive test suite for backend controllers, m...	yesterday
frontend	test: add comprehensive test suite for backend controllers, m...	yesterday
scratch	fixie	2 weeks ago
.gitignore	initialization	last month
PRD.md	for now	2 weeks ago
README.md	Add project details to README.md	1 minute ago
azure-pipelines.yml	Set up CI with Azure Pipelines	last week
vercel.json	vercel	2 weeks ago

README

About this project

Like 0

Overview

The Blog Management System is a web-based application designed to efficiently create, manage, and deliver blog content while improving user interaction and overall experience. It provides a centralized platform for administrators to handle blog operations and for users to access and engage with content seamlessly. The system is developed and managed using Azure DevOps, ensuring structured development, version control, and continuous integration and deployment.

Blog Content Management

About

bloggonut.vercel.app

Readme

Activity

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

Create a new release

Deployments 5

Production 1 minute ago

Preview yesterday

+ 3 deployments

Packages

No packages published

Publish your first package

Contributors 3

Naeha-S Naeha S

Copilot

NirthyThara

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.